Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

- Write out this design requirement in a single sentence, ensuring that it is clear, detailed, and measurable.
- Explain how this design requirement directly ties to your prior research.
- Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #1 (HIGH Priority)

- Must be capable of collecting microplastics as small as 1 mm, and as big as 5 mm
- This ties to our prior research in that microplastics are so incredibly small. If we just use a simple net, we won't capture the very small yet important microplastics

Design Requirement #2 (HIGH Priority)

- Must withstand the force of the water current
- Shouldn't be destroyed while being dragged through the water; can withstand both the force from the ROV, as well as external factors from either the ocean or the pool

Design Requirement #3 (HIGH Priority)

- No contamination from the constant use of the collection device.
- This was a problem in a previous design attempt for using pumps specifically.
- Other objects were getting into the pump, contaminating microplastic samples with other materials as well.

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Design Requirement #4 (HIGH Priority)

- Should match the buoyancy of the ROV or have neutral buoyancy.
- Cannot be dragging below or behind the ROV, which would lead to inaccurate or insufficient data.

Design Requirement #6 (LOW Priority)

- The collection device must be easy to remove and reattach from the ROV
- We learned that microplastics are incredibly small. This means that we must be thorough in removing samples from the device. This will be easier if we can detach it from the ROV

Design Requirement #7 (LOW Priority)

- Collection device must use as little plastic as possible
- We learned that storing samples in plastic contaminates samples with more microplastics.
- This is very low priority because we know that we will probably have to use many plastics based on our limited materials

Note on received feedback on this element: We received feedback on this element, but we were already past the timeframe of being able to change these design requirements. Regardless, here is what Josiah, from the Marine Biology class had to say: “Looks great to me, covers everything we went over in class when you all visited a few months ago,” and “One thing. What are you gonna do if marine life gets caught in your net?”